

to exit and reboot Raspberry Pi. Now type the command below to verify the

changed frequency (refer Fig. 4):

```
$ vcgencmd get config arm freq
```

[illegible]

Fig. 6: Firmware update

```

root@kali:~# nano /etc/cryptsetup.conf
=====
[opts]
expand_routes      Expand root partition to fill SD card
overcommit          Change overcommit
console_keymap     Set keyboard layout
change_pass         Change password for 'pi' user
change_locale       Set locale
change_timezone     Set timezone
memory_split        Change memory split
overcommit          Configure overcommitting
ask                Enable or disable ask server
boot_behavior       Boot behavior on boot
update             Try to upgrade dmccq-ioctl

<Enter>              <F10>
=====

```

Fig. 7: Raspi-config utility

P960 BIOS - Overclocking -
 Choose overclock preset:
 0 Basic 700MHz AMU, 1100MHz core, 4000MHz SDRAM, 0 overvolt A
 1 Robert 800MHz AMU, 1500MHz core, 4000MHz SDRAM, 0 overvolt A
 2 Medium 900MHz AMU, 1500MHz core, 4000MHz SDRAM, 2 overvolt A
 3 High 900MHz AMU, 1500MHz core, 4000MHz SDRAM, 4 overvolt A
 4 Turbo 1000MHz AMU, 800MHz core, 4000MHz SDRAM, 4 overvolt A
 <Back >Next

Fig. 8: Overclocking options

EarlyBoot menu

- BerryWebserver
- Berryterminal
- Debian Wheezy Raspbian
- Mandriva
- OpenELEC r1699
- OpenELEC r1791
- Puppy alpha 4
- Raspbian 2012.09
- Sugar

Buttons: Boot, Edit menu

Fig. 9: Berryboot menu

Note that the values mentioned above are the most stable ones, but I even managed to run the Raspberry Pi at 1 GHz with following configuration:

```
arm_freq=1000
core_freq=500
sdram_freq=500
over_voltage=6
force_turbo=0
```

Like this, you can change the frequency at which the processor is running, and you will notice the change when you navigate pages on Raspberry Pi itself. I noticed a great performance boost for XMBC Media Center.

If this method does not work for you, follow the other method of doing it as shown below. This method uses 'raspi-config' utility to overclock your Raspberry Pi.

Run 'sudo apt-get install raspberrypi* raspi-config' command in the terminal to update your firmware (refer Fig. 6).

Now type 'sudo raspi-config' in the terminal to open 'raspi-config' utility as shown in Fig. 7.

Select the 'overclock' option and press 'Enter.' You will see various overclocking options that you can select as shown in Fig. 8.

EFY note. Go beyond 800 MHz at your own risk. People have reported heating issues



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